

Supplement 2: Duration and Convexity as Derivatives

Calculus Derivations, Taylor Series and Redington Immunization (BKM Ch. 16)

Big picture: duration and convexity are not definitions to memorize. They are the first and second derivatives of the bond price function, divided by the price. Once you see that, the price-change formula is a Taylor series, and the classic actuarial immunization conditions are the first- and second-derivative tests for a minimum.

Source context: Companion to the slide deck "Duration, Convexity & Weighted Average Life" (Michigan Tech template) and to the notebook [FIN4000_Supp02_Notebook.ipynb](#), which runs every calculation below in SymPy and QuantLib. For the finance-side treatment (duration rules, calculator and spreadsheet steps, practice questions) see [Study Guide 5](#). BKM references are to *Investments*, 13th Edition, Chapter 16.

Prerequisites: the chain and power rules, Taylor series, and (for Section 6) the mean and variance of a discrete distribution.

What This Supplement Adds

This is enrichment for students in mathematics and actuarial science. It is not a numbered learning objective in BKM. By the end of it you should be able to:

- Derive modified duration and convexity by differentiating the price function.
 - Extend the derivation to k payments per year and to continuous compounding.
 - Show that the duration-convexity formula is a second-order Taylor expansion, and estimate its error with the third derivative.
 - Show that, under continuous compounding, convexity equals duration squared plus the variance of the payment times.
 - Prove Redington's immunization conditions with the second-derivative test.
 - Check each result symbolically and numerically in Python.
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1. Setup

For an option-free bond with fixed cash flows CF_t at times t (in years), the price is a function of the yield y (effective annual):

$$P(y) = \sum_t CF_t (1 + y)^{-t}$$

Write $PV_t = CF_t(1 + y)^{-t}$ for the present value of each payment and $w_t = PV_t/P$ for its weight. The weights are positive and sum to 1. Only y varies; the cash flows do not.

Running example: a 10-year bond with a 6% annual coupon and \$1,000 face value, priced at $y = 6\%$, so $P = \$1,000$.

2. Duration is the first derivative

Differentiate one term:

$$\frac{d}{dy}(1 + y)^{-t} = -t(1 + y)^{-t-1}$$

Sum over all payments and factor out one power of $1/(1+y)$:

$$\frac{dP}{dy} = - \sum_t t CF_t (1+y)^{-t-1} = - \frac{1}{1+y} \sum_t t PV_t$$

Divide by P and change sign:

$$-\frac{1}{P} \frac{dP}{dy} = \frac{1}{1+y} \sum_t t \frac{PV_t}{P} = \frac{D_{mac}}{1+y} = D^*$$

Here $D_{mac} = \sum_t t w_t$ is the Macaulay duration, a weighted average of the payment times (actuarial texts call it the discounted mean term, and call D^* the volatility). By the chain rule the same result reads:

$$D^* = - \frac{d \ln P}{dy}$$

so D^* is the percentage change in price per unit change in yield. The dollar version is the DV01:

$$DV01 = - \frac{dP}{dy} \times 0.0001 = P D^* \times 0.0001$$

Running example: $D_{mac} = 7.8017$ years, $D^* = 7.3601$, DV01 = \$0.7360 per \$1,000 face per basis point. A 1-point rise in yield gives a first-order estimate of -7.36% .

Two checks.

- Zero-coupon bond, $P = F(1+y)^{-T}$: $D^* = T/(1+y)$ and $D_{mac} = T$.
- Perpetuity, $P = c/y$, so $P' = -c/y^2$: $D^* = 1/y$ and $D_{mac} = (1+y)/y$. At $y = 5\%$, $D^* = 20$ and $D_{mac} = 21$.

3. Other compounding conventions

With k payments per year, $P(y) = \sum_t CF_t (1+y/k)^{-kt}$ and

$$\frac{d}{dy} (1+y/k)^{-kt} = -t (1+y/k)^{-kt-1} \Rightarrow D^* = \frac{D_{mac}}{1+y/k}$$

with t measured in years. Under continuous compounding with force of interest δ , $P(\delta) = \sum_t CF_t e^{-\delta t}$ and

$$-\frac{1}{P} \frac{dP}{d\delta} = \sum_t t w_t = D_{mac}$$

exactly. So Macaulay duration is the derivative with respect to the force of interest. The two results agree because $\delta = \ln(1+y)$ and $d\delta/dy = 1/(1+y)$:

$$\frac{dP}{dy} = \frac{dP}{d\delta} \cdot \frac{d\delta}{dy} = \frac{1}{1+y} \frac{dP}{d\delta}$$

4. Convexity is the second derivative

Differentiate dP/dy once more, using $\frac{d}{dy} [-t(1+y)^{-t-1}] = t(t+1)(1+y)^{-t-2}$:

$$\frac{d^2P}{dy^2} = \sum_t t(t+1) CF_t (1+y)^{-t-2}$$

Convexity is this second derivative divided by the price:

$$C = \frac{1}{P} \frac{d^2P}{dy^2} = \frac{1}{P(1+y)^2} \sum_t t(t+1) PV_t$$

Running example: $C = 69.7404$.

Every term in the sum is positive, so $P''(y) > 0$: the price-yield curve is convex for any bond with non-negative cash flows.

With k payments per year, the second derivative of $(1+y/k)^{-kt}$ is $\frac{t(kt+1)}{k}(1+y/k)^{-kt-2}$, so

$$C = \frac{1}{P(1+y/k)^2} \sum_t \frac{t(kt+1)}{k} PV_t$$

If you count time in periods, $n = kt$, this becomes $\sum n(n+1) PV_n / [P(1+y/k)^2 k^2]$. That is why [Study Guide 5](#) divides by 4 for semiannual bonds.

Two checks.

- Zero-coupon bond: $C = T(T+1)/(1+y)^2$. For $T = 10$ and $y = 6\%$: $D^* = 9.434$ and $C = 97.90$.
- Perpetuity: $P'' = 2c/y^3$, so $C = 2/y^2$. At 5%: $C = 800$.

5. The price-change formula is a Taylor series

Expand P around today's yield:

$$P(y + \Delta y) = P + P' \Delta y + \frac{1}{2} P'' (\Delta y)^2 + \frac{1}{6} P''' (\Delta y)^3 + \dots$$

Divide by P and substitute $P'/P = -D^*$ and $P''/P = C$:

$$\frac{\Delta P}{P} \approx -D^* \Delta y + \frac{1}{2} C (\Delta y)^2$$

The next term uses the third derivative:

$$\frac{P'''}{P} = -\frac{1}{P(1+y)^3} \sum_t t(t+1)(t+2) PV_t$$

By the Lagrange remainder, truncating after the second-order term leaves an error of $\frac{1}{6} P'''(\xi)(\Delta y)^3$ for some ξ between y and $y + \Delta y$. For the running example $P'''/P = -753.69$.

Yield change	Exact	Duration only	+ convexity	+ third order
+2 points	-13.4202%	-14.7202%	-13.3254%	-13.4259%
+1 point	-7.0236%	-7.3601%	-7.0114%	-7.0239%
-1 point	+7.7217%	+7.3601%	+7.7088%	+7.7214%
-2 points	+16.2218%	+14.7202%	+16.1150%	+16.2155%

- The first-order error is of order $(\Delta y)^2$ and has the same sign for rises and falls. The convex curve lies above its tangent line, so duration alone understates the price in both directions.
- The second-order error is of order $(\Delta y)^3$ and changes sign with Δy , because $P''' < 0$.

6. Convexity is duration squared plus variance

Treat the payment time T as a random variable that takes the value t with probability w_t . Then $D_{mac} = E[T]$. Under continuous compounding, differentiate twice with respect to δ :

$$\frac{1}{P} \frac{d^2 P}{d\delta^2} = \sum_t t^2 w_t = E[T^2] = D_{mac}^2 + \text{Var}(T)$$

At equal duration, a bond whose payments are more spread out in time has more convexity. That is why a barbell is more convex than a bullet.

Running example: $D_{mac} = 7.8017$, $\text{Var}(T) = 9.6922$, and $E[T^2] = 70.5586 = 7.8017^2 + 9.6922$.

To return to the effective-yield convexity of Section 4, differentiate $dP/dy = P_\delta/(1+y)$ once more:

$$\frac{d^2 P}{dy^2} = \frac{P_{\delta\delta} - P_\delta}{(1+y)^2} \Rightarrow C = \frac{D_{mac}^2 + \text{Var}(T) + D_{mac}}{(1+y)^2}$$

Check: $(70.5586 + 7.8017)/1.06^2 = 69.7404$, the same value as Section 4.

7. Redington immunization from derivatives

Let $A(y)$ and $L(y)$ be the present values of the asset and liability cash flows, and let $S(y) = A(y) - L(y)$ be the surplus. Assume a single parallel shift in yield. Redington (1952) showed that the surplus is protected at y_0 if:

1. $S(y_0) = 0$, so $A(y_0) = L(y_0)$.
2. $S'(y_0) = 0$, so $A'(y_0) = L'(y_0)$. With equal present values this means equal duration.
3. $S''(y_0) > 0$, so $A''(y_0) > L''(y_0)$. The assets must be more convex than the liabilities.

Then y_0 is a strict local minimum of S with $S(y_0) = 0$. By the second-derivative test and the Taylor expansion:

$$\Delta S \approx \frac{1}{2} S''(y_0) (\Delta y)^2 > 0$$

so any small move in yield, up or down, raises the surplus. When both sides have the same Macaulay duration, the second derivative reduces to a difference of variances:

$$S''(y_0) = \frac{L(y_0)}{(1+y_0)^2} [\text{Var}_A(T) - \text{Var}_L(T)]$$

Worked example. A liability of \$1,000 is due in 8 years and $y = 6\%$, so $L = \$627.41$. The assets are zero-coupon bonds maturing in years 5 and 12.

- Match duration: $5w + 12(1 - w) = 8$, so $w = 4/7$. Buy \$358.52 of the 5-year zero and \$268.89 of the 12-year zero.
- Convexity condition: $E[T^2]$ is $0.5714 \times 25 + 0.4286 \times 144 = 76$ for the assets and 64 for the liability, so the variance difference is 12.
- Result: $S''(6\%) = 627.41/1.06^2 \times 12 = 6,701$.

Yield	4.0%	5.0%	5.5%	6.0%	6.5%	7.0%	8.0%
Surplus S (\$)	1.60	0.37	0.09	0.00	0.08	0.31	1.13

At a 1-point move, $\frac{1}{2}(6701)(0.01)^2 = \0.34 , close to the table's \$0.37 and \$0.31.

Limits. The result holds for small, instantaneous, parallel shifts. Duration drifts as time passes, so the portfolio must be rebalanced. See BKM Section 16.3.

8. Checking with software

The notebook [FIN4000_Supp02_Notebook.ipynb](#) runs these checks. SymPy differentiates the price formula:

```
import sympy as sp
y = sp.symbols('y')
P = sum(60/(1+y)**t for t in range(1, 11)) + 1000/(1+y)**10
Dstar = -sp.diff(P, y) / P # D* = -(1/P) dP/dy
C = sp.diff(P, y, 2) / P # C = (1/P) d2P/dy2
Dstar.subs(y, 0.06), C.subs(y, 0.06) # 7.3601, 69.7404
```

QuantLib returns the same numbers from the bond object:

```
ql.BondFunctions.duration(bond, rate, ql.Duration.Modified) # 7.3601
ql.BondFunctions.convexity(bond, rate) # 69.7404
```

A central finite difference shows the derivative as a limit. With step h in the yield:

Step h	D^* (central difference)	C (second difference)
0.01	7.3727	69.8153
0.001	7.3602	69.7411
0.0001	7.3601	69.7404

9. Looking ahead: random rates

Everything above treats y as one number that is shifted deterministically. In derivatives and risk-management courses, the short rate r is a random process, for example the Vasicek model:

$$dr = a(b - r)dt + \sigma dW$$

Itô's lemma gives the price dynamics of a bond $P(t, r)$, where $\mu = a(b - r)$ is the drift of r :

$$dP = \left(P_t + \mu P_r + \frac{1}{2}\sigma^2 P_{rr}\right)dt + \sigma P_r dW$$

The term $\frac{1}{2}\sigma^2 P_{rr}$ is the convexity term again, now multiplied by the variance of the rate. For a zero-coupon bond under Vasicek, $-\partial \ln P / \partial r = (1 - e^{-ar})/a$, which is less than the maturity τ :

with $a = 0.1$ and $\tau = 10$ it is 6.32 years. Mean reversion shortens effective duration. As $a \rightarrow 0$ it approaches τ , which recovers “a zero's duration equals its maturity.”

10. Practice set

1. Derive D^* and C for a zero-coupon bond from $P(y) = F(1+y)^{-T}$. *Answer:* $D^* = T/(1+y)$ and $C = T(T+1)/(1+y)^2$.
2. For a perpetuity paying c a year, show that $D^* = 1/y$, $C = 2/y^2$ and $D_{mac} = (1+y)/y$. Evaluate at $y = 5\%$. *Answer:* 20, 800 and 21.
3. For the running example, use the third-order term to estimate the price change when yield rises 1 point. *Answer:* $-7.0114\% + \frac{1}{6}(-753.69)(0.01)^3 = -7.0239\%$; the exact change is -7.0236% .
4. Use $D^* = -d \ln P / dy$ to explain why D^* is the percentage price change per unit of yield. *Answer:* $d \ln P = dP/P$, so $d \ln P / dy$ is the relative price change per unit change in y .
5. A bullet pays \$1 at year 8. A barbell pays \$0.50 (in present value terms) at each of years 4 and 12. Both have $D_{mac} = 8$. Find $E[T^2]$ for each and the effective-yield convexity at $y = 6\%$. *Answer:* 64 and 80; $(64 + 8)/1.06^2 = 64.08$ and $(80 + 8)/1.06^2 = 78.32$. The variance difference is 16.
6. A liability of \$1,000 is due in 10 years at $y = 5\%$. Immunize it with zero-coupon bonds maturing in years 6 and 14. Find the weights and check the convexity condition. *Answer:* $6w + 14(1-w) = 10$, so $w = 0.5$: \$306.96 of each, since $L = \$613.91$. $E[T^2] = 0.5(36) + 0.5(196) = 116 > 100$, so the third condition holds.

11. Reading and open-source resources

- Bodie, Kane & Marcus, *Investments*, 13th ed., Chapter 16, especially Sections 16.1 (duration), 16.2 (convexity) and 16.3 (immunization).
- [Society of Actuaries, Using Duration and Convexity to Approximate Change in Present Value](#) (free PDF with derivation appendices).
- F. M. Redington, “Review of the Principles of Life-Office Valuations,” *Journal of the Institute of Actuaries* 78 (1952), 286–340.
- [QuantLib](#): `BondFunctions.duration`, `convexity` and `basisPointValue`, and the Vasicek and Hull-White short-rate models.
- [SymPy](#) for symbolic derivatives.
- [JuliaActuary](#): `ActuaryUtilities.jl` has duration and convexity, and is in this course's Julia environment.
- R package `FinancialMath`: `cf.analysis()` computes the same sums.
- [MIT OpenCourseWare 18.S096](#): Lecture 18 (Itô calculus), Lecture 21 (stochastic differential equations) and Lecture 24 (the HJM interest-rate model).

Key Terms

Make sure you can define each of these in your own words.

- ☐ price function $P(y)$
- ☐ modified duration D^*
- ☐ Macaulay duration
- ☐ volatility (actuarial)
- ☐ discounted mean term
- ☐ payment-time weights
- ☐ DV01
- ☐ convexity C

- ☐ Taylor series
 - ☐ Lagrange remainder
 - ☐ force of interest δ
 - ☐ variance of payment times
 - ☐ barbell and bullet
 - ☐ surplus function $S(y)$
 - ☐ Redington immunization
 - ☐ second-derivative test
 - ☐ short-rate model
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Key Equations

Modified duration

$$D^* = -\frac{1}{P} \frac{dP}{dy} = -\frac{d \ln P}{dy} = \frac{D_{mac}}{1+y}$$

Convexity

$$C = \frac{1}{P} \frac{d^2 P}{dy^2} = \frac{1}{P(1+y)^2} \sum_t t(t+1) PV_t$$

Price change

$$\frac{\Delta P}{P} \approx -D^* \Delta y + \frac{1}{2} C (\Delta y)^2$$

Convexity and dispersion (continuous compounding)

$$\frac{1}{P} \frac{d^2 P}{d\delta^2} = D_{mac}^2 + \text{Var}(T)$$

Redington conditions

$$S(y_0) = 0 \quad S'(y_0) = 0 \quad S''(y_0) > 0$$

Study guide supplement for the "Duration, Convexity & Weighted Average Life" slide deck. BKM section references are to Investments, 13th Edition, Chapter 16 (Managing Bond Portfolios). Numerical results were checked with SymPy and QuantLib.